

# AQA (Trilogy) Combined Science GCSE

## Unit 6.3 Particle Model - Answers

### Test (Levels 4 – 9)

|    |                                                                                                                                                                                                                                                                                           |                                              |
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| 1. |                                                                                                                                                                                                                                                                                           | (7 marks total)                              |
| a. | density = mass ÷ volume<br><b>accept</b> mass/volume                                                                                                                                                                                                                                      | (1 mark)                                     |
| b. | Two can be chosen from: <ul style="list-style-type: none"><li>• they are not all the same size</li><li>• ions/atoms/molecules are not solid circles</li><li>• there are no forces shown between the circles</li></ul>                                                                     | (1 mark)<br>(1 mark)<br>(1 mark)             |
| c. | The particles have more kinetic energy at higher temperatures<br>... this means there is an increase in (average) speed<br>... which means the collisions with the container wall happen more often<br>... so the wall of the container has a larger force applied meaning pressure rises | (1 mark)<br>(1 mark)<br>(1 mark)<br>(1 mark) |
| 2. |                                                                                                                                                                                                                                                                                           | (4 marks total)                              |
| a. | kilograms per metre cubed, kg/m <sup>3</sup>                                                                                                                                                                                                                                              | (1 mark)                                     |
| b. | <b>In this order only:</b><br>constantly<br>kinetic                                                                                                                                                                                                                                       | (1 mark)<br>(1 mark)                         |
| c. | pressure: increases/rises                                                                                                                                                                                                                                                                 | (1 mark)                                     |

| 3. |                                                                                                                                                                                                                                                | (14 marks total)                                                             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| a. | the amount of energy needed to change 1kg of liquid into 1kg of gas<br>at the same temperature                                                                                                                                                 | (1 mark)<br>(1 mark)                                                         |
| b. | gas particles have random motion ( <b>accept move freely</b> )<br>and are far away from each other<br><b>Whereas</b> , solid particles are tightly packed ( <b>accept in regular patterns</b> )<br>and vibrate around positions that are fixed | (1 mark)<br>(1 mark)<br>(1 mark)                                             |
| c. | 47 460 or $4.746 \times 10^4$ (J)<br><b>Accept</b> 47 000 or $4.7 \times 10^4$<br>$0.021 \times 2.26 \times 10^6$ gets 1 mark only                                                                                                             | (2 marks) or<br>(2 marks) or<br>(1 mark)                                     |
| d. | A is melting point<br>C is boiling point<br><br>AB state change is solid to liquid ( <b>accept melting</b> )<br>At a steady temperature<br><br>BC is liquid state and the temperature is increasing<br>Reaching boiling point at C             | (1 mark)<br>(1 mark)<br><br>(1 mark)<br>(1 mark)<br><br>(1 mark)<br>(1 mark) |

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| 4. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (8 marks total)                              |
| a. | latent heat of fusion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (1 mark)                                     |
| b. | <b>Choose one from:</b> <ul style="list-style-type: none"> <li>current of the element</li> <li>mass of ice</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | (1 mark)<br>(1 mark)                         |
| c. | <b>The answer should use good grammar, full sentences and have valid points made:</b> <ul style="list-style-type: none"> <li>Temperature increases and heats the wires</li> <li>Increase caused by current in the wires</li> <li>Ice is solid and has a regular arrangement of particles</li> <li>Particles vibrate around fixed points</li> <li>Ice melts as the internal energy increases to melting point</li> <li>Particles gain more energy as the temperature increases</li> <li>Particles vibrate more quickly</li> <li>This increased vibration breaks bonds between particles</li> <li>Particles become irregularly arranged as in a liquid</li> <li>Particles have more random movement</li> </ul> | (6 marks)                                    |
| 5. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | (11 marks total)                             |
| a. | random motion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (1 mark)                                     |
| b. | to change the state of water, energy is needed to break bonds                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (1 mark)<br>(1 mark)                         |
| c. | <b>Three points from:</b> <ul style="list-style-type: none"> <li>heating will increase the speed of the particles</li> <li>as they have more kinetic energy</li> <li>the gas will increase in temperature</li> <li>which is proportional to the kinetic energy</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    | (1 mark)<br>(1 mark)<br>(1 mark)<br>(1 mark) |
| d. | $1000 = m \div 3.1 \times 10^{-5}$<br>$m = 3.1 \times 10^{-5} \times 1000$<br>$m = 0.031 \text{ (kg)}$<br><br><b>using <math>E = m \times L</math></b><br><br>$E = 0.031 \times 2.26 \times 10^6 \text{ (or } 0.031 \times 2\,260\,000)$<br><br>$E = 70\,060 \text{ (J)}$<br><br><b>Full 5 marks if 70 060 (J) with no working out shown</b>                                                                                                                                                                                                                                                                                                                                                                 | (5 marks)                                    |

| 6. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | (6 marks total)                                                                 |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| a. | <p><b>Make sure the object fits into a measuring cylinder or Eureka can easily:</b></p> <ul style="list-style-type: none"> <li>- Measure the object's mass</li> <li>- Put a known volume of water into the measuring cylinder or Eureka can that will submerge the object but not overflow</li> <li>- Insert the object and measure the new volume</li> <li>- The volume of the object is the amount of displacement of the volume of water</li> <li>- Density is calculated using <math>\text{mass} \div \text{volume}</math></li> </ul> | <p>(1 mark)</p> <p>(1 mark)</p> <p>(1 mark)</p> <p>(1 mark)</p> <p>(1 mark)</p> |
| b. | 4cm <sup>3</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | (1 mark)                                                                        |